MCS Saturday 7th-SHSAT Math Homework

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Grade 7 SHSAT Math · CAT Digital Format

Percent Decision Lab

Session 3 Homework Activity Sheet · All-New Questions

Name _____ **Date** _____

Completion Check	Core	Transfer	Total
Accuracy + denominator evidence + CAT commit routine	_____ /18	_____ /8	_____ /26

A CAT Commit Protocol

5 prompts · 7 min

Practice the forward-only decision routine. Give observable actions rather than general advice.

Read the ask | **Mark the start** | **Predict size** | **Verify form** | **Commit once**

A1 · Commit cue Recall

You have solved an item and checked the requested quantity. What exact action ends your work on that screen?

Name the action and what happens next.

A2 · Adaptive meaning Explain

Your next item feels easier than the previous one. Explain why one item's difficulty does **not** determine your final result.

Refer to ongoing score estimation, not one question.

A3 · Typed form Apply

Your work gives **0.285**, and the screen asks for the numerical value of the percent. What should you enter if the percent sign is not permitted?

Enter one value

Enter only the requested numerical value.

A4 · Denominator check Reason

Before advancing on a percent-change item, what phrase in the story should you connect to the denominator?

State the phrase and why it controls the denominator.

A5 · Pace decision Defend

After about two minutes, your setup is complete but your arithmetic is unfinished. Write a two-sentence rule for making the best supported response and moving forward without panic.

Sentence 1: finish or estimate. Sentence 2: verify form and commit.

B Benchmark Warm-Up

6 questions · 9 min

Write a 10%, 5%, 1%, or equivalent-fraction benchmark before calculating.

10% = divide by 10 · 5% = half of 10% · 1% = divide by 100 · Predict before computing

B1 Level 1

Find **10%** of **730**.

Show the decimal-place shift

Typed entry _____

B2 Level 1

Find **5%** of **340**.

Start from 10%

Typed entry _____

B3 Level 1

Find **1%** of **8,400**.

Name the place-value move

Typed entry _____

B4 Level 2

Find **15%** of **260**.

Combine two benchmark parts

Typed entry _____

B5 Level 2

Which is the best estimate of **32%** of **190**?

A. 6

B. 19

C. 61

D. 158

Use 30% as an anchor

B6 Level 2

Write **42.5%** as both a decimal and a fraction in lowest terms.

Percent → decimal → fraction

Decimal; fraction _____

C Core Accuracy Set

12 questions · 24 min

Treat each card as one forward-only screen: identify the whole or original, predict, solve, scan the response form, and commit.

Amount = percent × whole · Percent = part ÷ whole · Change ÷ original · Magnitude is nonnegative

C1 Level 2

Find **24%** of **350**.

Decompose 24% or multiply

One number _____

☐ I checked that my answer is less than 350.

C2 Level 3

Find **42.5%** of **160**.

Use 40%, 2.5%, or an equivalent fraction

One number _____

☐ I compared my result with half of 160.

C3 Level 2

96 is what percent of **320**?

Identify part and whole before dividing

Percent number only _____

☐ I converted the quotient to a percent.

C4 Level 3

A quantity rises from **175** to **217**. Find the percent increase.

Change ÷ original × 100

Percent number only _____

☐ I used 175 as the denominator.

C5 Level 3

A reservoir level falls from **560** centimeters to **434** centimeters. Find the percent decrease.

Estimate before exact division

Percent number only _____

☐ I reported a positive magnitude.

C6 Level 2

A value increases by **27** from an original value of **180**. What is the percent increase?

The change is already given

Percent number only _____

☐ I did not subtract 27 again.

C Core Accuracy Set (continued)

12 questions · 24 min

Treat each card as one forward-only screen: identify the whole or original, predict, solve, scan the response form, and commit.

Amount = percent × whole · Percent = part ÷ whole · Change ÷ original · Magnitude is nonnegative

C7 · Multi-select Level 3

Select **all** expressions equal to **35%** of **240**.

A. 0.35 × 240

B. (35/100) × 240

C. 30% of 240 + 5% of 240

D. 240 ÷ 0.35

Verify each selected expression

☐ I evaluated every option.

C8 · Direction Level 3

A value changes from **120** to **150**, then returns to **120**. Which change has the greater percent magnitude?

A. The increase

B. The decrease

C. They are equal

D. Not enough information

Show both original values

☐ I used a new launchpad for each change.

C9 · Reverse percent Level 4

72 is **30%** of what number?

Write 0.30 × whole = 72

Whole number _____

☐ I checked 30% of my answer.

C10 · Error analysis Level 4

To find the percent increase from **240** to **300**, Maya computes **60 ÷ 300**. Identify and correct her first wrong decision.

Name the wrong denominator and replace it

☐ My explanation names the original value.

C11 · Context Level 3

At a community event, **45%** of **440** attendees arrived before noon. How many arrived at noon or later?

Find the requested group, not the given group

People _____

☐ I answered the complement asked for.

C12 · Two changes Level 4

A count moves from **200** to **230**, then to **207**. Compare the percent increase in the first change with the percent decrease in the second.

Calculate each percent from its own original

Comparison sentence _____

☐ I used 200 first and 230 second.

D SHSAT Transfer Set

8 questions · 20 min

Model the full situation before choosing an operation. The final line should contain only the requested quantity.

D1 · Part of a part Level 3

A museum sold **680** tickets. Of the tickets, **35%** were sold online. One fourth of the online tickets were student tickets. How many online student tickets were sold?

Whole → online part → student part

Tickets _____

D2 · Percent then fraction Level 3

In a survey of **250** students, **72%** chose an after-school activity. Of those students, **5/9** chose a sport. How many surveyed students chose a sport?

Find the subgroup in two stages

Students _____

D3 · Geometry transfer Level 3

A rectangle’s width stays fixed while its length increases from **25** units to **31** units. By what percent does the rectangle’s area increase?

Use area structure; a width value is unnecessary

Percent number only _____

D4 · Ratio transfer Level 4

The angles of a triangle are in the ratio **3:5:7**. The smallest angle is what percent of the largest angle?

The common scale factor may cancel

Percent number only _____

D SHSAT Transfer Set (continued)

8 questions · 20 min

Model the full situation before choosing an operation. The final line should contain only the requested quantity.

D5 · Statistics transfer Level 3

The mean of five quiz scores is **64**. A sixth score of **88** is added. By what percent does the mean increase?

Recover the old total, then find the new mean

Percent number only _____

D6 · Multi-select Level 3

Select **all** values that are greater than **37%** but less than **0.4**.

A. 3/8

B. 0.385

C. 39.5%

D. 2/5

Convert to one representation

D7 · Two categories Level 4

A store has **360** pens. Blue pens are **40%** of the total. Black pens are **25%** of the pens that are not blue. How many pens are neither blue nor black?

Track the changing whole carefully

Pens _____

D8 · Claim analysis Level 4

A student says, “If a population increases **20%** one year and decreases **20%** the next year, it returns to its starting value.” Test the claim using a convenient starting value and explain the denominator shift.

Example + conclusion + two original values

E One-Screen Rehearsal

3 decisions · 5 min

Imagine the Next button locks the current item. State the final check you would perform in each situation.

E1. Your answer for **18%** of a positive number is greater than the original number. What structural check should interrupt your commit?

State the expected size relationship.

E2. Your percent decrease is negative, but the prompt asks for the magnitude of the change. What belongs in the response field?

Separate direction from magnitude.

E3. The prompt asks how many students, but your field contains “34%.” Identify the mismatch and state your next action before advancing.

Name the requested quantity and correct the form.

F Process Debrief

Evidence, not impressions

Complete this before checking solutions. Focus on decisions you can repeat under forward-only conditions.

- ☐ I named the requested quantity before calculating.
- ☐ I identified the whole for every percent-of problem.
- ☐ I found the change before dividing on percent-change items.
- ☐ I divided each change by its own original value.
- ☐ I predicted whether the answer should be smaller or larger.
- ☐ I used benchmarks to audit exact arithmetic.
- ☐ I entered only the value and form requested.
- ☐ I committed after one evidence-based final check.

Timing record

Record your time by section and identify where your pace changed.

A–B _____ **C** _____

D–E _____ **Total** _____

Confidence audit

List two items you would most want to revisit on a non-adaptive test. Then explain the verification you used instead.

Item numbers + substitute verification

G Error Log

Up to 5 corrections

Use one primary type: Concept, Procedure, Reading, Arithmetic, Format, or Wrong Quantity.

#	Problem	First wrong decision	Primary type	Next-time interruption cue
1				
2				
3				
4				
5				

H Transfer Reflection

Complete sentences

1. The benchmark that made one exact calculation easier was…

2. The words that tell me which value is the original are…

3. One typed-entry mismatch I can now catch is…

4. My five-second percent-change commit question will be…